INDIE MAPS

Say it naturally. Navigate intelligently.

I. Project Overview

Indie Maps is a multilingual, voice-first conversational navigation assistant that transforms natural spoken instructions including regional languages, Hinglish, and code-mixed expressions into real-world navigation actions. Users can discover places, set destinations, add multiple stops, and modify journeys through conversation without relying on rigid search queries or manual route configuration.

II. The Problem

Traditional navigation applications rely on structured searches, exact place names, and manual route management. However, people naturally describe locations through landmarks, refer to businesses by category, combine multiple instructions, and switch between languages.

Understanding that a tea stall near a landmark is the intended destination not the landmark itself or distinguishing a food stop from a final destination requires contextual and geographical reasoning that conventional search workflows do not inherently provide.

III. Our Solution

Indie Maps bridges the gap between natural human conversation and structured navigation. Users simply speak their intentions in their own language. The application interprets the request, identifies geographical relationships, discovers relevant locations, and converts the instruction into executable navigation actions.

IV. Key Capabilities

Multilingual Voice Interaction: Processes spoken instructions across configured Indian languages and code-mixed expressions using Gnani AI's speech technologies.

Landmark-Aware Discovery: Separates business categories from geographical anchors, finding tea stalls near Bangla Sahib rather than mistakenly navigating to the landmark itself.

Multi-Intent Planning: Extracts multiple operations from one instruction, distinguishing final destinations from intermediate stops.

Contextual Journey Management: Supports conversational follow-ups, alternative selection, and route modifications while preserving existing destination and stop relationships.

Real-World Navigation: Integrates Google Places API (New) and Routes API v2 to discover locations and calculate updated routes.

V. How the Technology Works

Voice Input → Gnani AI Speech-to-Text → Gemini Intent Interpretation → Structured Navigation Actions → Journey Transaction Engine → Google Places API & Routes API → Updated Route → Gnani AI Text-to-Speech.

Gnani AI powers speech recognition and spoken responses, while Gemini interprets transcribed instructions and extracts structured navigation intent. The Journey Transaction Engine executes these actions while maintaining journey context. Google Places enables location discovery, and Routes API v2 calculates resulting routes. A rule-based fallback supports selected command patterns.

VI. Innovation and Technical Differentiation

Indie Maps moves beyond voice-enabled map search by translating natural conversation into geographically meaningful, state-aware navigation operations.

Its key innovation is understanding not just which places a user mentions, but what role each place plays in the journey. It distinguishes landmarks used as search references from actual destinations, and intermediate businesses from final destinations. It also separates landmark-anchored discovery from route-relative detour searches.

Beyond individual commands, Indie Maps maintains journey state across interactions, preserving destination and stop relationships as users refine their routes. Its multi-intent execution further decomposes a single instruction into multiple operations for example, navigating to a destination while adding both a food stop and a fuel station along the way.

Combining multilingual speech, contextual intent interpretation, geographical validation, and consistent multi-operation execution transforms navigation from isolated searches into conversational journey management.

VII. Technology Stack

Frontend: React, TypeScript, Vite, Tailwind CSS

Backend: Node.js, Express, WebSockets

AI & Speech: Google Gemini, Gnani AI / Vachana Speech-to-Text and Timbre Text-to-Speech

Mapping: Google Maps JavaScript API, Places API (New), Routes API v2

VIII. Expected Impact

Indie Maps aims to make navigation more intuitive, inclusive, and hands-free by reducing dependence on exact search terms and repetitive manual interactions. It demonstrates how multilingual speech technology and contextual AI can transform conventional utility applications into systems that understand and act on natural human instructions.

INDIE MAPS — From natural conversation to intelligent navigation.